(DMR) Common Task Framework for UCT Processing: Auto Labeling & Data Storage

USSF Space Ops Command

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West Lafayette, IN Rockies

In this project, you'll work directly with Space Operations Command (SpOC) and the SDA TAP Lab to help shape the future of space data science. Your team will design tools to build benchmark datasets—essential for training and testing machine learning models used in Space Domain Awareness (SDA). These tools will support the processing of Uncorrelated Tracks (UCTs), which are unknown or untracked objects in space.

Using open-source data and working with subject matter experts, you'll automate the labeling, cleaning, and organizing of real space data (like satellite maneuvers and launch events). Then, you'll help define how algorithm performance should be measured—so results can be fairly compared across developers using a Common Task Framework.

This is an exciting opportunity for students interested in artificial intelligence, space operations, or software development to be part of building the next generation of tools for national security and space traffic management.

— See Details —

Fall mentor time: Thursday: 3:30 PM Eastern
Fall lab time: Tuesday: 3:30 PM Eastern
Spring mentor time: Thursday: 3:30 PM Eastern
Spring lab time: Tuesday: 3:30 PM Eastern
Industry: Government
Requirements: U.S. citizens only

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